

L05 Chihuahua or Muffin with CNN

CNN Architecture and Model Performance

The goal of this lab was to repeat the Chihuahua-or-muffin classification task using a convolutional neural network instead of the fully connected neural network from Lab 04. I used the same small dataset: 120 training images and 30 validation images. The images were resized to 64 by 64 pixels, normalized, and loaded in batches of 32. The training pipeline also used random horizontal flips and rotations up to 10 degrees. This data augmentation did not create genuinely new specimens, but it exposed the model to reasonable variations in orientation and made it less likely to memorize the exact training images.

The CNN contained three convolutional blocks. The first convolution changed the three RGB channels into 32 feature maps, the second increased this to 64, and the third increased it to 128. Each convolution was followed by ReLU and max pooling. The pooling steps reduced the spatial dimensions from 64 by 64 to 32 by 32, then 16 by 16, and finally 8 by 8. The result was flattened and passed through a 512-neuron layer, dropout set to 0.5, and a final two-neuron output layer. The model had 4,289,090 trainable parameters, with most concentrated in the fully connected portion.

Unlike the earlier multi-layer perceptron, the CNN does not immediately treat the image as one long list of unrelated pixels. Its filters examine local neighborhoods and reuse the same filter across the image, allowing the network to learn edges, textures, and shapes while retaining spatial relationships. Module 05 describes this as looking locally, sharing filters, and stacking layers into a feature hierarchy.

The CNN trained for 10 epochs with the Adam optimizer and a learning rate of 0.001. Training accuracy increased from 57.50% in the first epoch to 94.17% in the tenth. Validation accuracy increased from 56.67% to 96.67%, with the best validation score first appearing at epoch five and returning in epochs seven through ten. The final model correctly classified 29 of 30 validation images. All 17 Chihuahua images were classified correctly. The only error was a top-down blueberry muffin that was labeled as a Chihuahua. Its round outline and dark blueberry spots may have resembled the general arrangement of a face, which is a concrete example of how a model can learn visual shortcuts rather than the concept a person believes it is learning.

Comparison, Challenges, and Experimentation

The previous traditional neural network reached 86.67% validation accuracy, or 26 correct predictions out of 30, after three epochs. The CNN reached 96.67%, reducing the validation errors from four images to one. This was a meaningful improvement on the same dataset, although the validation set is too small to treat a one-image difference as a stable estimate of real-world performance. Each validation image changes the reported accuracy by 3.33 percentage points. A larger independent test set would be needed before concluding that the model reliably generalizes.

The CNN required more computation because it ran for 10 epochs and performed convolution and pooling before classification. Its saved progress bars show a run under a

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minute on CPU, but neither notebook recorded formal total timing. Since the MLP ran for only three epochs, an exact speed ratio would be misleading; a fair comparison would use the same epochs, device, and timing code.

The main technical challenge was the environment setup. Upgrading PyTorch before Torchvision briefly created a version conflict, resolved by upgrading both. The DataLoader also warned that four workers exceeded the runtime's recommended two. Training completed, but setting `num_workers` to two would be cleaner. I selected 64-by-64 images because they balanced detail with manageable memory and runtime.

The main configuration I evaluated used 10 epochs, a 0.001 learning rate, three convolutional layers, and 0.5 dropout. The epoch results still showed the effect of training duration: validation accuracy reached 96.67% at epoch five, dropped to 80.00% at epoch six, and recovered. A controlled follow-up should change only one variable, such as lowering the learning rate to 0.0005 or comparing five epochs with ten. This would separate a real parameter effect from normal variation in a small dataset.

Real World Applications and Personal Reflection

The same approach can support microscopy, medical imaging, manufacturing inspection, agriculture, and laboratory quality control. In biological work, a CNN could screen cell images, classify colony morphology, identify tissue patterns, or flag unusual assay images. Transfer learning would often be more practical than training from scratch because early CNN layers learn reusable edges and textures while deeper layers become task-specific.

The ethical issue is not the muffin error itself, but the same failure pattern inside a high-stakes application. High average accuracy can hide failure on a class, image source, demographic group, instrument, or laboratory condition. Training data may not represent the deployment population, and confident predictions can still be wrong. Image datasets also raise privacy and consent concerns. NIST emphasizes validity, reliability, transparency, privacy, explainability, and managed bias as parts of trustworthy AI (Tabassi, 2023). That requires documented data sources, testing across relevant conditions, monitoring after deployment, and human review when errors have serious consequences.

Overall, this lab made the advantage of convolution much clearer than the lecture alone. The MLP could solve much of the task, but the CNN used the structure of the image instead of discarding it at the beginning. The improvement from 86.67% to 96.67% matched the conceptual argument from the course: local filters and feature hierarchies are better suited to visual patterns. At the same time, the one misclassified muffin and the very small validation set kept the result in perspective. The model worked well for a classroom demonstration, but a polished accuracy number is not the same as validation for real use. That distinction is directly relevant to biological and laboratory applications, where an automated result should be treated as evidence that must be checked, not as an unquestionable answer.

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References

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